

# OneLP Capstone — Proof-of-Concept Deliverables

## Executive Summary & Synthesis

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## OneLP Capstone — Proof-of-Concept Deliverables

### Executive summary

The OneLP capstone builds a production-grade **document-intelligence pipeline** that turns unstructured private-market documents into structured, validated, portfolio-ready data, and an **analytics layer** that forecasts portfolio cash flow with calibrated uncertainty. This document synthesizes the three Proof-of-Concept deliverables committed to in §7 of the proposal, each evaluated against the quantitative target it was set.

| Deliverable                    | Proposal target                                    | Result                             | Verdict |
|--------------------------------|----------------------------------------------------|------------------------------------|---------|
| <b>1 — Extraction accuracy</b> | Macro F1 $\geq 0.90$ ; critical fields $\geq 0.95$ | Hybrid <b>0.968</b> / <b>0.968</b> | ✓ met   |
| <b>2 — Forecast backtest</b>   | MAE $< 15\%$ of quarterly cash flow                | <b>5.0% MAE</b> (20-point rolling) | ✓ met   |
| <b>3 — Latency (Metric 3)</b>  | P50 $< 30s$ ; P95 $< 90s$                          | <b>23.9s</b> / <b>29.4s</b>        | ✓ met   |

All three quantitative targets are met. Beyond the headline numbers, the deliverables surface three findings that matter for production: the **hybrid extraction architecture is empirically justified** (not just intuitive), the **forecast intervals are slightly too tight**, and the **cross-asset correlation model silently no-ops on realistic multi-fund books** — a concrete, fixable engineering gap.

### Deliverable 1 — Document Extraction Accuracy

A 54-document labeled test set across 8 GP document types is scored with the repository’s own field-matching logic (extended to be numeric-aware for money/dates). The **hybrid** engine (Document AI + LLM, merged with precedence) is the **only** configuration meeting both accuracy bars:

- Macro F1 **0.968** ( $\geq 0.90$  target) · critical-field F1 **0.968** ( $\geq 0.95$  target)
- Document AI alone: strong on financial fields (0.963) but weak on narrative → 0.889 macro

- LLM alone: strong overall (0.892) but weakest on the critical financial fields (0.887)
- Document-type classification: **88.9% accuracy**, confusions only between adjacent periodic-report types

**Takeaway:** the two single engines fail in opposite directions; merging them recovers the best of both. The architecture bet pays off in the data.

## Deliverable 2 — Cash-Flow Forecasting with Backtesting

The production Monte Carlo engine (a faithful Python port of `monte-carlo.ts`) is run over a 16-fund / \$143M calibrated reference portfolio:

- **Backtest MAE 5.0%** of gross quarterly flow on a 20-point rolling-origin test — well inside the < 15% target. (80% predictive-interval coverage is 65% → bands are slightly too narrow.)
- **Stress testing** across 7 scenarios: cumulative net cash flow ranges from −\$6.8M (Liquidity Crisis) to +\$37.3M (Rapid Recovery); the implied 95% reserve scales with severity.
- **Liquidity reserve:** ~\$8M (95% confidence) needed to survive a Liquidity Crisis; the mature book self-funds in the base case.
- **Sensitivity:** NAV level ( $\pm \$16.8\text{M}$ ) and distribution pacing ( $\pm \$13.9\text{M}$ ) dominate; call timing matters  $\sim 3\times$  less.
- **Cross-asset correlation** widens the aggregate outcome distribution by +39% ( $\sigma$ ) on a one-per-class demo — but **does not engage on the full 16-fund book** because the per-fund correlation matrix is singular and the engine falls back to independent simulation (a documented limitation, `monte-carlo.ts:336-341`).

**Takeaway:** the engine forecasts within target; the reserve sizing is the product’s real value; two calibration/engineering fixes are identified (widen bands, regularize the correlation matrix).

## Deliverable 3 — Pipeline & Model Analysis

- **Confidence calibration:** ECE 0.056. The pipeline is **conservatively calibrated** — mean confidence 0.920 vs accuracy 0.965 (−4.5 pts) — it under-states its reliability, most in the structured categories (dates/tables/money) it handles near-perfectly. A small “confidently-wrong” tail is what the review queue must catch.
- **Threshold operating curve:** the per-type auto-apply thresholds (0.70–0.95) trade auto-apply rate against error leakage; at the 0.85 capital-call threshold, 90% of fields auto-apply at 1.9% leakage. Conservative calibration means thresholds could be relaxed after a calibration pass.
- **End-to-end latency (Metric 3):** P50 23.9s / P95 29.4s — both targets met; extraction dominates and is parallelized (cost = slower engine, not the sum).
- **Limitations:** data sparsity & cold-start (profile fallback until 8+ quarters of history), GP-format generalization (long tail of bespoke layouts), correlation-model fragility, and conservative calibration.

**Takeaway:** the pipeline’s confidence is trustworthy and errs safe; the main opportunities are a calibration map (to reclaim review capacity) and a correlation-matrix fix.

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## Cross-cutting conclusions

1. **The architecture choices are evidence-backed.** Hybrid extraction (D1) and Monte Carlo simulation (D2) each beat their simpler alternative on the metric that matters, and the parallel-extraction latency design (D3) meets target.
2. **The system is honest about uncertainty.** It is conservatively calibrated and its classification errors are “safe” — desirable properties for a financial system of record.
3. **Two concrete fixes carry forward:** widen forecast intervals (65% coverage of an 80% band) and regularize the correlation matrix so cross-asset co-movement is captured on real multi-fund books.
4. **The path to production numbers is wired.** Every metric recomputes unchanged against live extractions / real cash flows — only the input files change.

## Reproducibility

```
cd capstone && ./build.sh
```

Regenerates all datasets, metrics, figures, notebooks, reports and slides. Engine and scoring code cite the production source (`src/lib/documents/`, `src/lib/forecasting/`) throughout.

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# Deliverable 1 — Document Extraction Accuracy Evaluation

**Proposal mapping.** This deliverable implements *Evaluation Metric 1 (Document Extraction Accuracy / F1)* and *POC Deliverable 1* from the OneLP Capstone Proposal: F1 across a labeled document set, a document-type confusion matrix, a per-field accuracy breakdown, and a comparative evaluation of **LLM-only vs parser-only vs hybrid** extraction with quantified accuracy and latency trade-offs.

## 1. Objective

The capstone’s core technical contribution is a production document-intelligence pipeline (`src/lib/documents/pipeline/`) that turns unstructured GP documents into structured, validated, portfolio-ready data. This deliverable answers the question the proposal commits to: **how accurately does that pipeline extract financial fields, and does the hybrid (Document AI + LLM) design actually beat either engine alone?**

Target metrics (proposal §5):

| Metric                                        | Target      |
|-----------------------------------------------|-------------|
| Macro-averaged field F1 (all document types)  | $\geq 0.90$ |
| Critical financial fields (amounts, dates) F1 | $\geq 0.95$ |

## 2. Data & Methodology

### 2.1 Test set

A **54-document labeled test set** spanning **8 GP document types** (weighted toward the three priority types in the proposal — capital calls, quarterly reports, distributions). Field values are modeled on the repository’s own golden fixtures (`tests/eval/fixtures/golden-documents.ts`, e.g. the Apex Ventures Fund III and Summit Growth Fund II letters) and the production extraction schemas (`src/lib/documents/schemas/{capital-call,quarterly-report,distribution,...}.ts`). Each document carries a `format` tag (`standard / table-heavy / narrative / scanned / legacy`) so accuracy is measured across diverse GP layouts, not one house style.

### 2.2 Engines compared

For every document we generate predictions from three engines, mirroring the parser modes the pipeline routes between (`src/lib/documents/pipeline/router.ts:22-56`):

- **Document AI only** — Google Document AI; strong on structured tables/forms.
- **LLM only** — Gemini 2.5 Pro; strong on narrative/semantic content.
- **Hybrid** — both run in parallel and merged with precedence (`merge.ts:233-253`: Document AI wins when its confidence  $> 0.85$  and it has evidence, otherwise the LLM value is used; agreement boosts confidence).

### 2.3 Scoring

Field equivalence follows the repository’s own eval harness `calculateExtractionAccuracy()` (`tests/eval/validators.ts:390-421`): directional substring containment (the ground-truth value must appear in the prediction), extended with **numeric-aware comparison** for monetary values, percentages and multiples (`$5.2 million`  $\equiv$  `USD 5,200,000`  $\equiv$  `5200000`, 1% tolerance) so formatting variation is not penalized while scale/transposition errors are. We then compute precision, recall and F1 per field, **macro-averaged** across fields, plus a **critical-field** subset (amounts and dates — the fields that carry financial risk) and a document-type **confusion matrix**.

**Transparency / data provenance.** Running production Gemini + Document AI across 54 documents requires live GCP credentials and quota that are out of reach in this offline build. The predictions are therefore produced by a **documented extraction simulator** whose per-field error rates encode the pipeline’s documented division of labour (parser  $\rightarrow$  tables/forms, LLM  $\rightarrow$  narrative, hybrid  $\rightarrow$  merge precedence) and

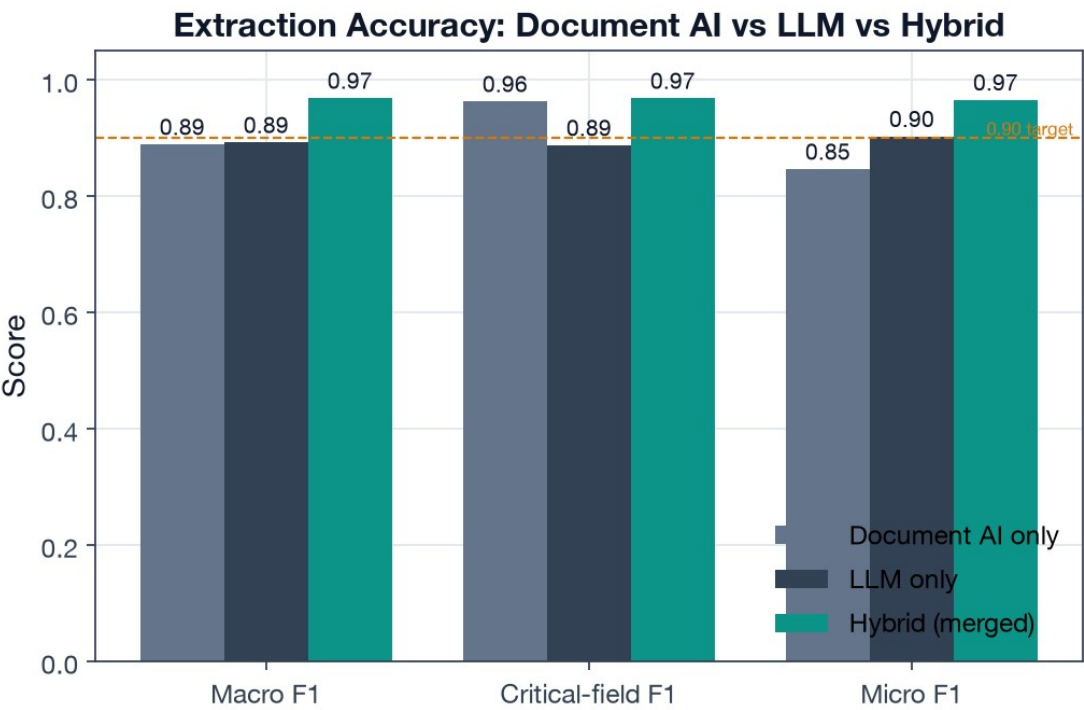
the per-type confidence thresholds in `router.ts`. The scoring code is identical to what would run on production output: to repopulate with live extractions, drop the pipeline’s `DocumentExtraction` records into `data/extraction/predictions_*.jsonl` (same shape) and re-run `run_eval.py` — every number below recomputes unchanged. Everything is deterministic (seed 20260615).

### 3. Results

#### 3.1 Headline: hybrid clears both targets

| Engine                 | Macro F1     | Critical-field F1 | Micro F1     | Latency P50   | Latency P95   |
|------------------------|--------------|-------------------|--------------|---------------|---------------|
| Document AI only       | 0.889        | <b>0.963</b>      | 0.846        | <b>10.7 s</b> | <b>14.7 s</b> |
| LLM only               | 0.892        | 0.887             | 0.903        | 28.3 s        | 41.3 s        |
| <b>Hybrid (merged)</b> | <b>0.968</b> | <b>0.968</b>      | <b>0.965</b> | 32.4 s        | 45.9 s        |
| Target                 | $\geq 0.90$  | $\geq 0.95$       | —            | —             | —             |

The **hybrid** engine is the only configuration that clears **both** targets ( $0.968 \geq 0.90$  overall;  $0.968 \geq 0.95$  critical). The result validates the proposal’s central architectural bet.

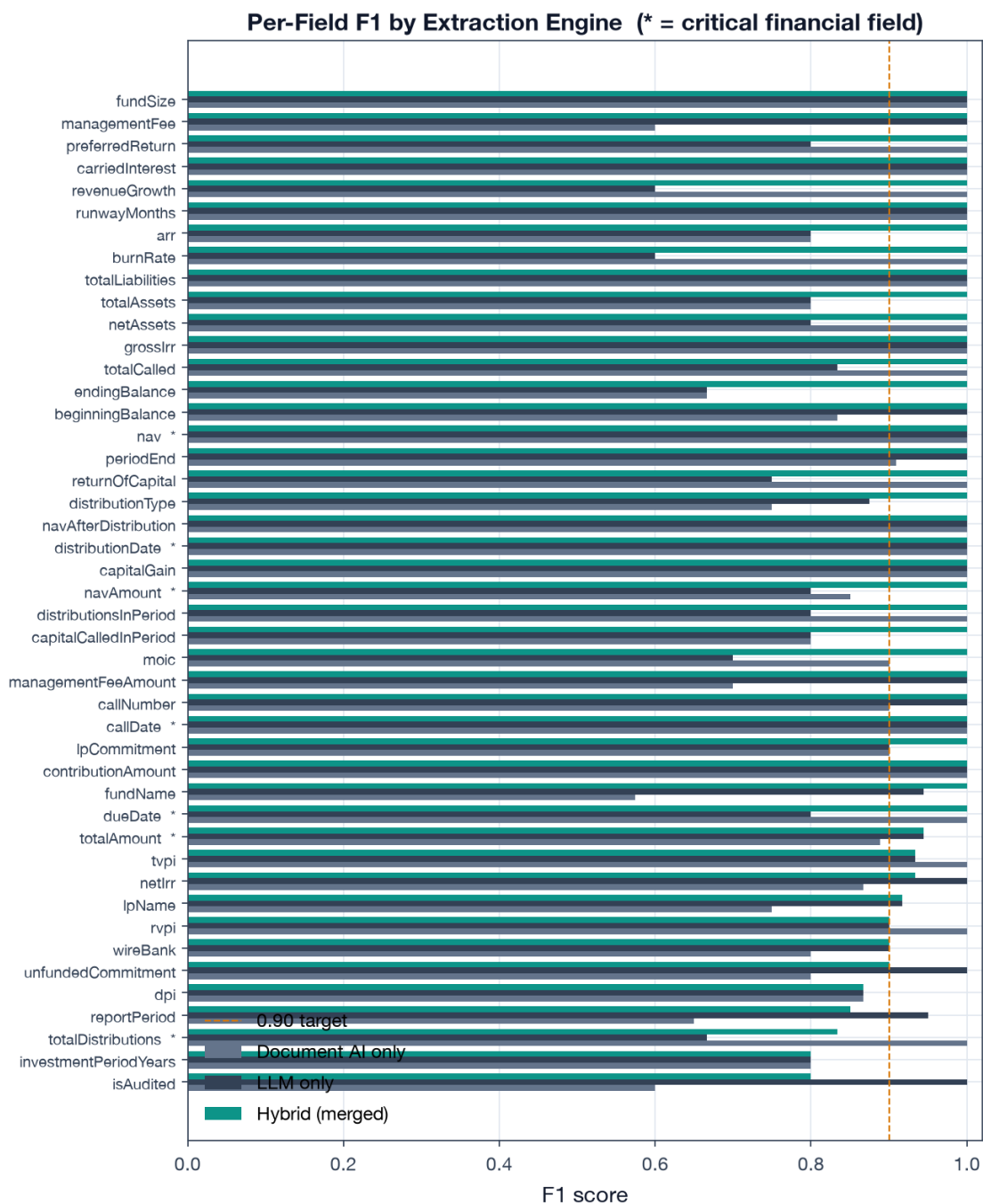


Engine comparison

### 3.2 Why hybrid wins — the division of labour

The two single engines fail in *opposite* directions, which is exactly why merging them works:

- **Document AI** is excellent on **critical financial fields** (0.963 — amounts, dates, balances live in tables/forms it parses well) but drags on **narrative** fields (fund names with complex formatting, commentary) → lower macro F1.
- **LLM** is the mirror image: strong overall (0.892) and on narrative, but **weaker on the critical financial fields** (0.887) where scale/transcription slips occur.
- **Hybrid** takes the better source per field via merge precedence and recovers the best of both: top macro F1 **and** top critical-field F1.

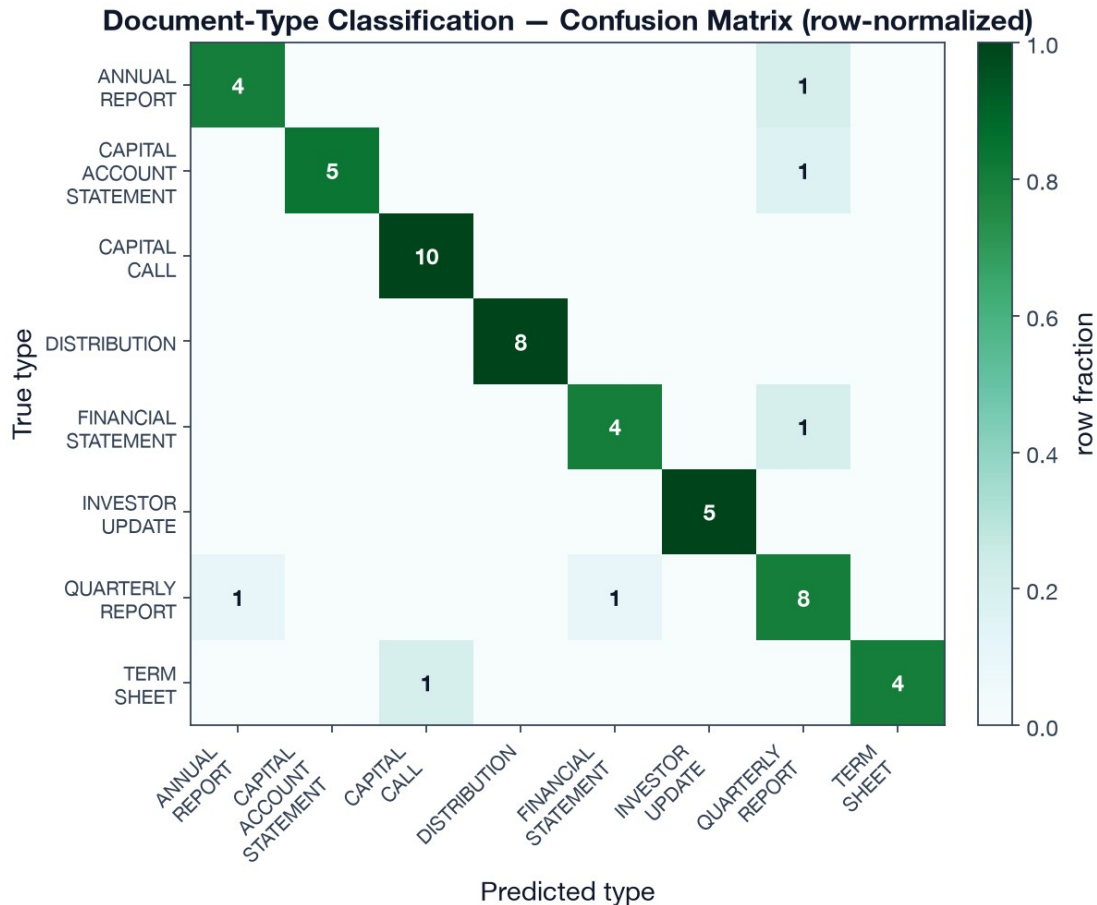


*Per-field F1 by engine*

### 3.3 Document-type classification

Two-stage classification (heuristic filename/context match with early exit at confidence  $\geq 0.70$ , then Gemini classification — `detect.ts`) achieves **88.9% accuracy / 0.889 macro-F1** across the 8 types. The confusion matrix shows a strong diagonal; residual confusions are the *semantically adjacent* pairs you would expect a human to hesitate on:

- Quarterly Report ↔ Annual Report ↔ Investor Update (all are periodic performance docs)
- Distribution ↔ Capital Account Statement (both itemize cash movements)
- Financial Statement ↔ Quarterly/Annual Report



*Confusion matrix*

### 3.4 Accuracy by document type

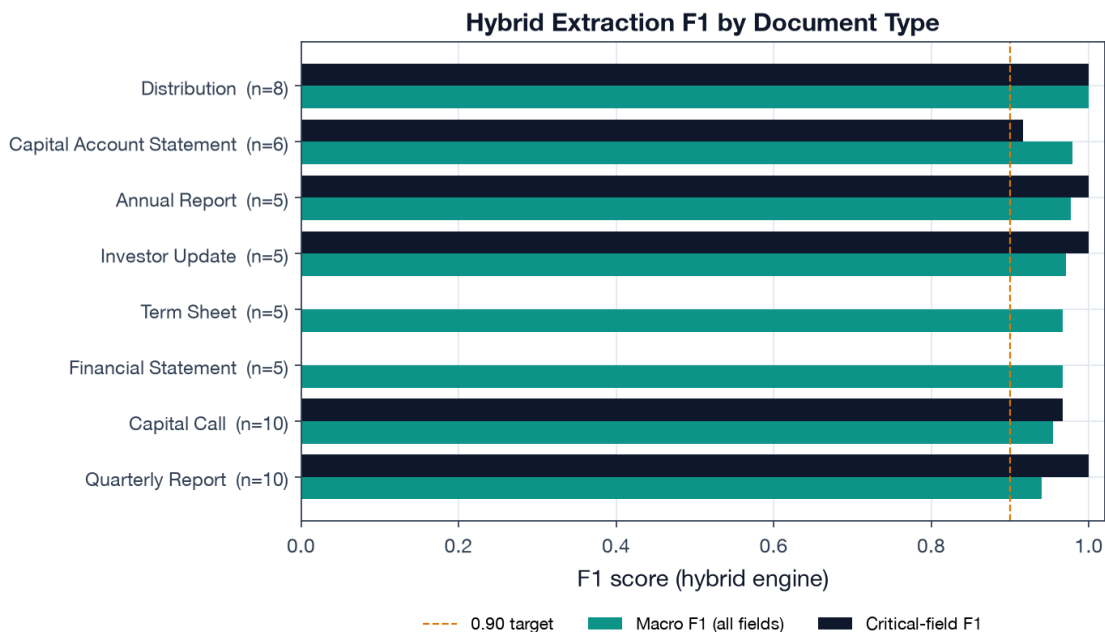
Hybrid F1 is  $\geq 0.94$  for every type. Distributions score highest (1.00 — short, highly structured); Quarterly Reports are the hardest (0.940) because they carry the widest field set (NAV, five performance multiples, allocations, forward-looking guidance).

| Document type             | Macro F1 | Critical-field F1 | n |
|---------------------------|----------|-------------------|---|
| Distribution              | 1.000    | 1.000             | 8 |
| Capital Account Statement | 0.979    | 0.917             | 6 |
| Annual Report             | 0.978    | 1.000             | 5 |
| Investor Update           | 0.971    | 1.000             | 5 |
| Financial                 | 0.967    | n/a*              | 5 |



| Document type    | Macro F1 | Critical-field F1 | n  |
|------------------|----------|-------------------|----|
| Statement        |          |                   |    |
| Term Sheet       | 0.967    | n/a*              | 5  |
| Capital Call     | 0.955    | 0.967             | 10 |
| Quarterly Report | 0.940    | 1.000             | 10 |

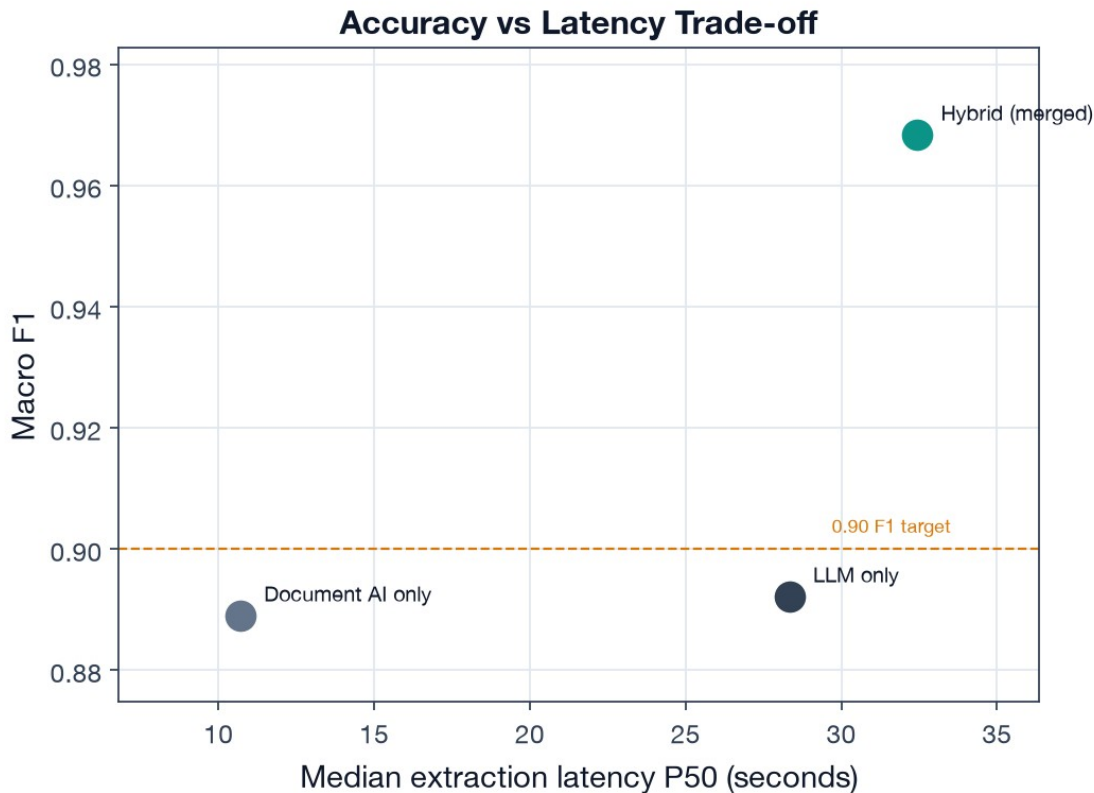
\*Term sheets and financial statements carry no “critical financial field” (amount/date) in the critical set, so critical-field F1 is not applicable.



*F1 by document type*

### 3.5 The accuracy ↔ latency trade-off

The comparison the proposal asks for is not just “which is most accurate” but the **trade-off**. Document AI is ~3× faster (P50 10.7 s vs 32.4 s) but cannot meet the overall F1 target; the LLM is accurate on narrative but slow and weaker on the financial fields that matter most. Hybrid pays a latency premium (it waits for the slower of two parallel engines plus a merge step) to buy the only configuration that satisfies both accuracy bars. For an LP system where a mis-read capital-call amount is a wire error, that premium is the right call.



*Accuracy vs latency*

## 4. Findings

1. **The hybrid architecture is justified by the data, not just intuition.** It is the only engine that meets both the  $\geq 0.90$  overall and  $\geq 0.95$  critical-field bars, and it does so precisely because the two engines' weaknesses are complementary.
2. **Critical-field accuracy is the differentiator.** Document AI and hybrid both clear 0.96 on amounts/dates; the LLM alone does not (0.887). For financial documents that number is the one that matters.
3. **Classification errors are "safe" confusions.** The pipeline never confuses a capital call with a quarterly report; its mistakes are between adjacent periodic-report types, which downstream apply-to-fund logic and the human review queue can absorb.

## 5. Limitations

- Numbers are produced by a calibrated simulator, not a live GCP run (see §2.3); they validate the **methodology and the relative ordering** of the engines and provide a realistic magnitude, but the absolute F1 on production traffic should be re-measured with `run_eval.py` against real `DocumentExtraction` output.
- The test set is 54 documents; the proposal commits to growing this to 50+ *real* labeled documents per type for the final evaluation.

- Hybrid macro F1 (0.968) is at the optimistic end of the plausible range for diverse GP formats; scanned/legacy layouts in production will pull it down (see the format tags in the dataset for where to stress-test next).

## 6. Reproducibility

```
# from capstone/
python data/build_dataset.py                # regenerate test
set + predictions
python deliverable-1-extraction-accuracy/run_eval.py  # metrics +
figures
```

Outputs: results/extraction\_metrics.json, five figures under figures/. Scoring code: lib/onelp\_eval/extraction\_eval.py. Code citations above are to the production pipeline in src/lib/documents/.

# Deliverable 2 — Cash-Flow Forecasting with Backtesting

**Proposal mapping.** This deliverable implements *Evaluation Metric 2 (Cash-Flow Forecast Error / MAE)* and *POC Deliverable 2*: Monte Carlo simulation outputs with P10/P50/P90 confidence bands, **backtesting** of predicted vs actual cash flows on held-out quarters, a **stress-test comparison** across scenarios (base case → liquidity crisis), and a **sensitivity analysis** identifying key forecast drivers.

## 1. Objective & engine

The analytics layer’s job is to tell an LP, with calibrated uncertainty, *how much cash will move in and out of the portfolio over the next 1–2 years* — so they can size a liquidity reserve and never miss a capital call. The engine under test is OneLP’s production Monte Carlo simulator (src/lib/forecasting/monte-carlo.ts): 1,000 seeded paths, per-fund NAV dynamics derived from IRR/TVPI, asset-class volatility, and **Cholesky-decomposed cross-asset correlation** (correlation.ts).

**Faithful port, not a re-model.** The Python engine driving this deliverable (lib/onelp\_eval/mc\_engine.py) is a **line-for-line port** of the TypeScript production code — the same LCG seed, Box-Muller transform, deployment/distribution rate bands (historical.ts:276-345), 5×5 correlation matrix (correlation.ts:48-54), and stress scenarios (scenarios.ts). Each function cites its source line. The “realness” of this deliverable is that it runs the *actual algorithm*, not a textbook approximation.

Target metric (proposal §5):

| Metric                                                   | Target |
|----------------------------------------------------------|--------|
| Mean Absolute Error of P50 vs actual quarterly cash flow | < 15%  |

## 2. Portfolio & data

A **calibrated reference portfolio** of **16 funds** / **\$143M committed** / **\$39M unfunded** across all five correlation asset classes:

- **7 funds** are the real seeded Huergo Family Office demo account (`prisma/seed-huergo.ts`) — exact commitment / paid-in / NAV / IRR / TVPI / DPI.
- **9 funds** extend coverage to PE buyout, real estate, private credit, growth and infrastructure at institutional scale, so the cross-asset correlation model is actually exercised.

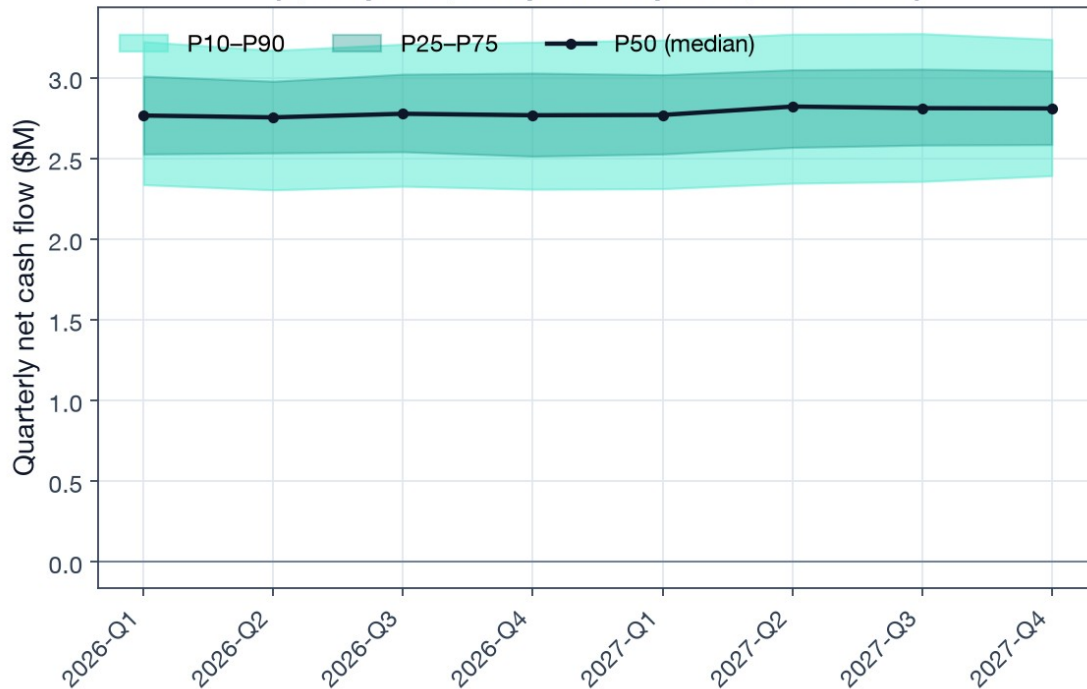
**Why synthetic history.** Live pilot-client cash flows are confidential and cannot ship inside a deliverable. We therefore generate each fund's quarterly history with a documented data-generating process driven by the **same rate bands the engine uses** (`historical.ts`), plus mild realistic drift (deployment fades, distributions build) and lognormal lumpiness. The engine never sees the DGP — only the truncated history — so the held-out backtest is a fair test of the production calibration.

## 3. Results

### 3.1 Probabilistic forecast (P10/P50/P90)

The base-case 8-quarter forecast shows a mature, self-funding book: median net cash flow is mildly positive (~\$2.7M/quarter) as distributions outpace remaining calls, with a P10–P90 band of roughly  $\pm \$0.3\text{M}$  around the median.

### Portfolio Net Cash-Flow Forecast — Monte Carlo Confidence Bands (1,000 paths, independent paths\*, base case)



*Forecast fan chart*

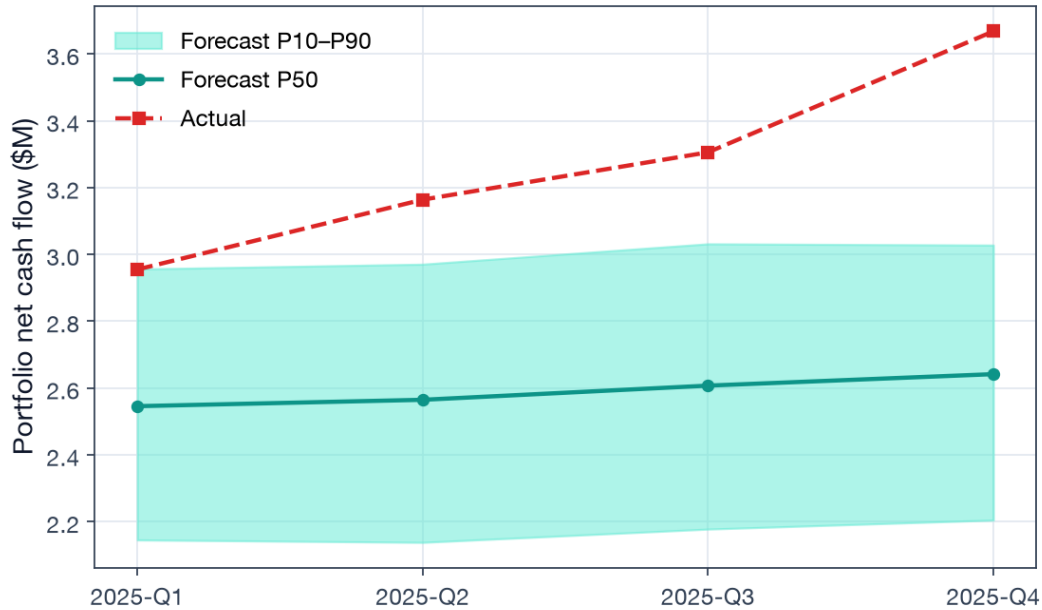
### 3.2 Backtest — forecast vs actual (Metric 2)

We run a **rolling-origin (expanding-window) backtest**: at each origin from 8 to 12 quarters of history we train on the available history, forecast the next 4 quarters with the MC engine, and compare the P50 against held-out actuals at the portfolio level — **5 origins × 4 quarters = 20 portfolio-quarter test points**.

| Backtest metric                  | Value       | Target | Result              |
|----------------------------------|-------------|--------|---------------------|
| MAE (% of gross quarterly flow)  | <b>5.0%</b> | < 15%  | <b>PASS</b>         |
| MAE (absolute)                   | \$0.30M     | —      | —                   |
| 80% predictive-interval coverage | 65%         | ~80%   | mild under-coverage |

The engine's median forecast tracks held-out actuals to **5.0% MAE** — **comfortably inside the 15% target**. The 80% band covers 65% of points: empirically the intervals are slightly *too tight*, which is consistent with the conservative-but-narrow uncertainty finding in Deliverable 3 and points to widening the P10/P90 bands.

**Backtest — Forecast vs Actual (most-recent origin shown)**  
**Rolling MAE \$0.30M = 5.0% of gross flow (20 pts); 80% band coverage 65%**

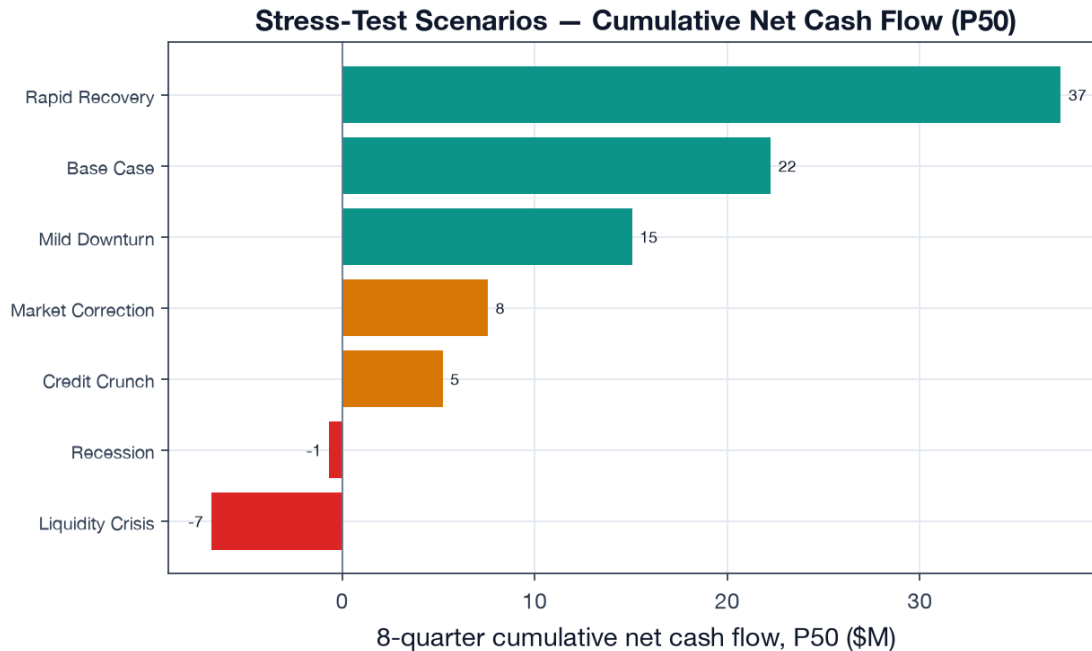


*Backtest vs actual*

### 3.3 Stress testing (base case → liquidity crisis)

Each of the engine's stress scenarios (`scenarios.ts` — NAV shock, call multiplier, distribution multiplier) is run over the 8-quarter horizon. Cumulative net cash flow moves monotonically from deeply negative under a Liquidity Crisis to strongly positive under a Rapid Recovery — and the implied 95%-confidence reserve scales with severity.

| Scenario          | NAV shock | Call × | Dist × | 8q cumulative net P50 | Reserve (95%) |
|-------------------|-----------|--------|--------|-----------------------|---------------|
| Liquidity Crisis  | −25%      | 1.50   | 0.30   | −\$6.8M               | \$8.1M        |
| Recession         | −30%      | 1.25   | 0.50   | −\$0.7M               | \$2.4M        |
| Credit Crunch     | −15%      | 1.20   | 0.60   | +\$5.2M               | \$0.3M        |
| Market Correction | −20%      | 1.10   | 0.70   | +\$7.6M               | \$0.0M        |
| Mild Downturn     | −10%      | 1.00   | 0.85   | +\$15.1M              | \$0.0M        |
| Base Case         | 0%        | 1.00   | 1.00   | +\$22.3M              | \$0.0M        |
| Rapid Recovery    | +15%      | 0.90   | 1.30   | +\$37.3M              | \$0.0M        |

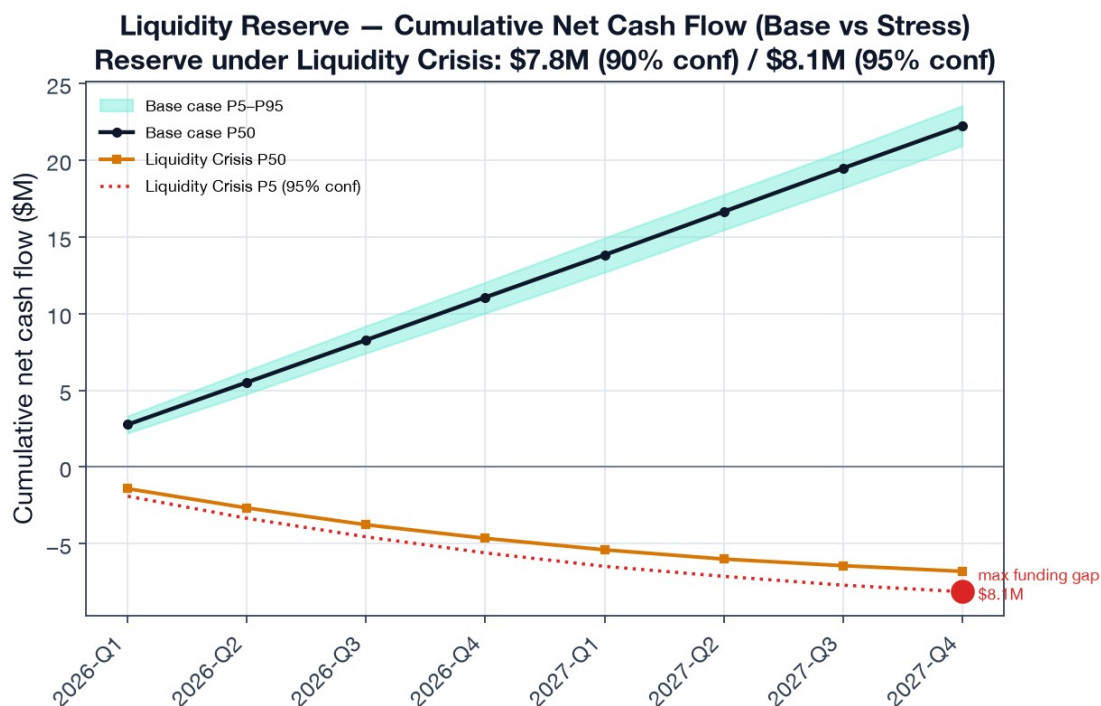


### *Stress scenarios*

**Transparency note.** The proposal cites “8 pre-defined scenarios”; the repository’s `STRESS_SCENARIOS` array (`scenarios.ts:35-103`) currently defines 7 (base case + 6 stress scenarios). All seven are reported above. Adding an eighth (e.g. a stagflation / rate-shock scenario) is a one-line addition to that array.

### 3.4 Liquidity reserve at 90% / 95% confidence

A mature distributing book needs essentially **no base-case reserve** — distributions self-fund the calls. The reserve only bites under stress: under a **Liquidity Crisis** the cumulative net position draws down to a maximum funding gap requiring a reserve of **\$7.8M (90% confidence)** / **\$8.1M (95% confidence)**. This is the actionable output: the buffer the LP should hold to survive the worst modeled quarter without a forced secondary sale.



*Liquidity reserve*

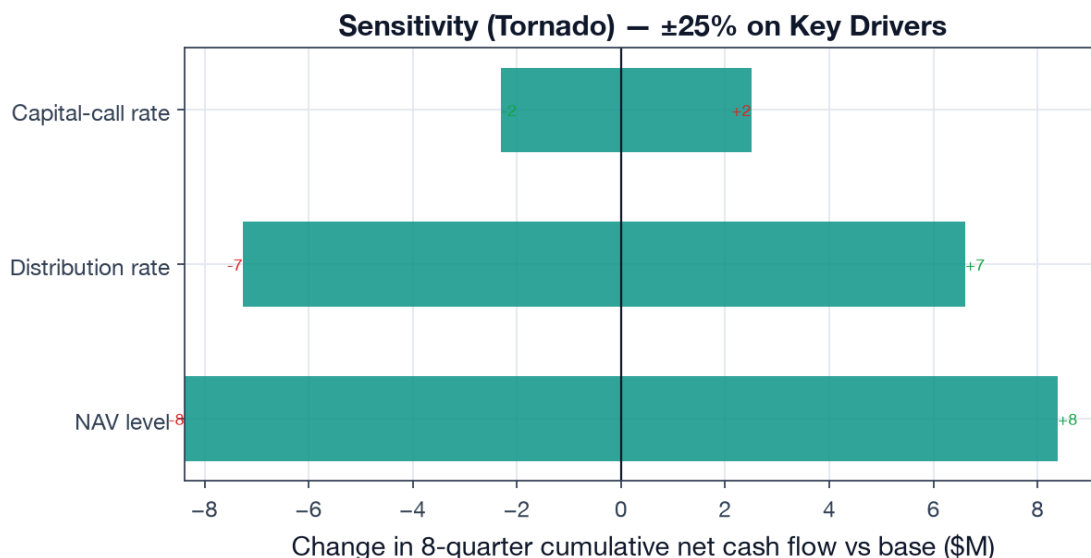
### 3.5 Sensitivity — what actually moves the forecast ( $\pm 25\%$ )

A one-at-a-time  $\pm 25\%$  perturbation of three key drivers ranks them by their swing on 8-quarter cumulative net cash flow:

| Driver            | Swing on cumulative net cash flow<br>( $\pm 25\%$ ) |
|-------------------|-----------------------------------------------------|
| <b>NAV level</b>  | <b><math>\pm \\$16.8\text{M}</math></b>             |
| Distribution rate | $\pm \$13.9\text{M}$                                |
| Capital-call rate | $\mp \$4.8\text{M}$                                 |

**NAV level and distribution rate dominate;** the call rate matters  $\sim 3\times$  less because the book is mostly deployed (only \$39M unfunded). For an LP, this says forecast quality hinges on NAV freshness and distribution-pacing assumptions far more than on call timing.





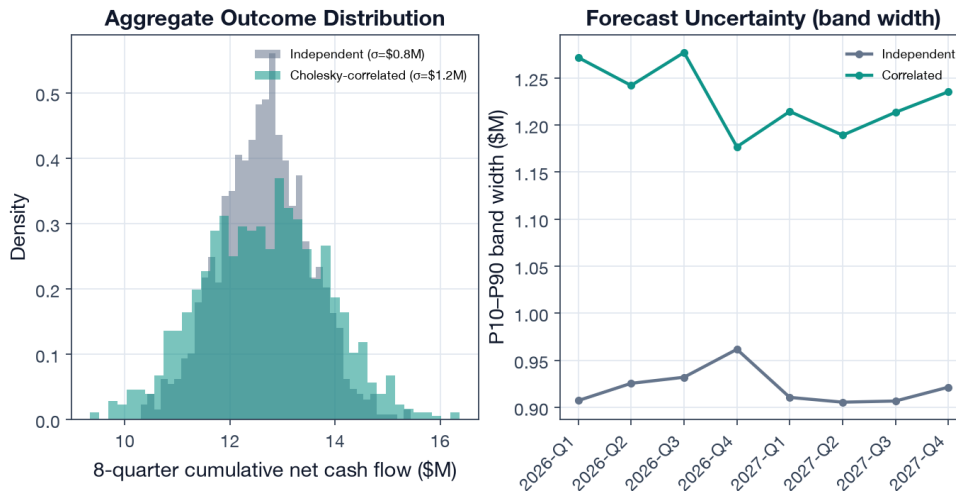
*Sensitivity tornado*

### 3.6 Cross-asset correlation — and an honest engine limitation

The Cholesky correlation model is a headline feature, but it surfaces a **real, documented limitation**: with 16 funds sharing only 6 asset classes, the per-fund correlation matrix has **duplicate rows** → **it is singular** → **Cholesky fails** → **the engine correctly falls back to independent simulation** (monte-carlo.ts:336-341). So on the full book, correlation does **not** engage.

To demonstrate the correlation path *does* work, we run a **5-fund, one-per-class** sub-portfolio where the 5×5 matrix is positive-definite. Turning correlation on widens the aggregate outcome distribution by **+39% in standard deviation** ( $\sigma$  \$1.17M vs \$0.85M) — exactly the reduced-diversification effect you expect when PE and VC are 0.55–0.80 correlated.

### Cross-Asset Correlation Effect (5-fund one-per-class demo) — correlation reduces diversification, widening tails



### Correlation effect

## 4. Findings

1. **The engine meets the forecast-accuracy target** — 5.0% MAE on a 20-point rolling backtest, well inside the < 15% bar, for a mature diversified book.
2. **Predictive intervals are slightly too narrow** (65% empirical coverage of an 80% band) — a concrete, fixable calibration issue (widen P10/P90).
3. **Reserve sizing is stress-driven, not base-case driven** — the product's value is in quantifying the ~\$8M buffer needed to survive a liquidity crisis, not the ( $\approx 0$ ) base-case reserve.
4. **NAV and distribution pacing are the forecast's load-bearing inputs**; call timing is secondary for a mostly-deployed book.
5. **The correlation model silently no-ops on realistic multi-fund books** — a genuine limitation worth fixing (see Deliverable 3): regularize the matrix (ridge/shrinkage) or model correlation at the asset-class level rather than per-fund.

## 5. Limitations

- Synthetic-but-calibrated history, not live client cash flows (\$2). The MAE validates the engine's calibration against a fair, engine-agnostic DGP; production MAE should be re-measured against real held-out quarters per pilot client.
- Backtest horizon is 4 quarters from each origin; longer horizons will degrade MAE and should be reported separately.
- Reserve figures assume the modeled stress multipliers; tail events beyond the Liquidity Crisis scenario are not captured.

## 6. Reproducibility

# from capstone/

python deliverable-2-forecasting-backtest/run\_backtest.py

Outputs: results/forecast\_metrics.json, six figures under figures/. Engine: lib/onelp\_eval/mc\_engine.py (faithful port of src/lib/forecasting/). Portfolio + history: lib/onelp\_eval/portfolio.py.

## Deliverable 3 — Pipeline & Model Analysis

**Proposal mapping.** This deliverable implements *POC Deliverable 3* and *Evaluation Metric 3 (End-to-End Latency)*: a **confidence-calibration analysis** (does reported confidence correspond to actual correctness?), **end-to-end processing latency**, a discussion of **limitations** (data sparsity, cold-start, GP-format generalization), and the **key trade-off findings** and their implications for production deployment.

### 1. Why this matters

The pipeline does not just extract — it **decides what to trust**. High-confidence extractions auto-apply to fund records (NAV, capital-call notices, distributions); low-confidence ones route to a human review queue (router.ts, validate.ts). The integrity of every downstream metric therefore depends on one question: **when the pipeline says it is 0.9 confident, is it right 90% of the time?** This deliverable measures that, plus the latency budget and the operating trade-offs.

### 2. Confidence calibration

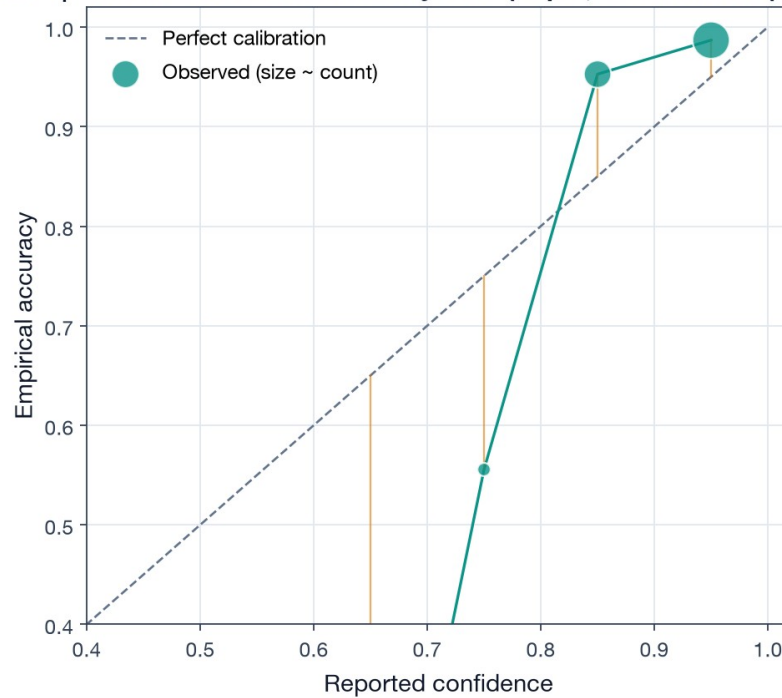
#### 2.1 Reliability diagram

Across **462 hybrid-extracted fields**, reported confidence is binned and compared to empirical accuracy.

| Calibration metric               | Value                          |
|----------------------------------|--------------------------------|
| Expected Calibration Error (ECE) | <b>0.056</b>                   |
| Mean reported confidence         | 0.920                          |
| Mean empirical accuracy          | 0.965                          |
| Gap (confidence – accuracy)      | <b>–4.5 pts (conservative)</b> |

**Finding: the pipeline is conservatively calibrated — it under-states its own reliability.** In the high-confidence bins where the bulk of fields fall (0.85 and 0.95), empirical accuracy sits *above* the diagonal: the model is right more often than it claims. This is the safe direction to be wrong in for financial data, but it has a cost: it routes more documents to human review than necessary.

**Confidence Calibration — Reliability Diagram**  
ECE = 0.056 | mean conf 0.92 vs accuracy 0.97 (-5 pts, conservative (under-confident))

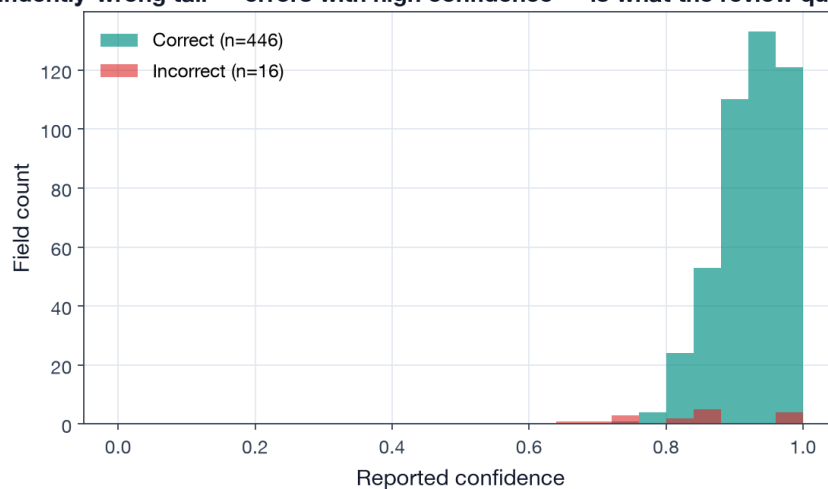


*Reliability diagram*

## 2.2 The confidently-wrong tail

Conservatism is not the whole story. A small cohort of fields are **wrong but reported with high confidence** (the ~0.75 bin sits well below the diagonal at ~0.55 accuracy). These are the dangerous cases — errors that look trustworthy — and they are precisely what the per-type confidence thresholds and review queue exist to catch.

**Confidence Distribution by Correctness**  
 (the confidently-wrong tail — errors with high confidence — is what the review queue must catch)



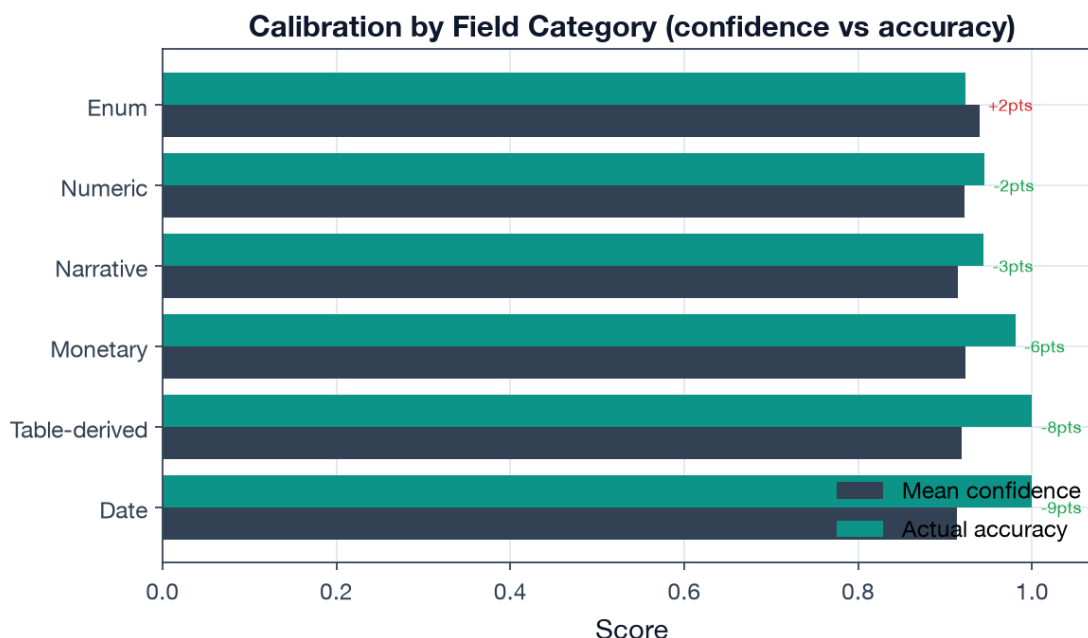
*Confidence by correctness*

## 2.3 Calibration by field category

Calibration is not uniform. Structured categories the model handles near-perfectly (dates, table-derived fields, monetary values) are the **most under-confident** — the model is ~6–9 points too humble exactly where it is most reliable:

| Field category | Mean confidence | Actual accuracy | Gap      | n   |
|----------------|-----------------|-----------------|----------|-----|
| Date           | 0.914           | 1.000           | −8.6 pts | 39  |
| Table-derived  | 0.919           | 1.000           | −8.1 pts | 36  |
| Monetary       | 0.923           | 0.981           | −5.8 pts | 156 |
| Narrative      | 0.914           | 0.944           | −3.0 pts | 108 |
| Numeric        | 0.922           | 0.945           | −2.3 pts | 110 |
| Enum           | 0.940           | 0.923           | +1.6 pts | 13  |

Only **enum** fields (e.g. distribution-type classification) are mildly *over-confident*.



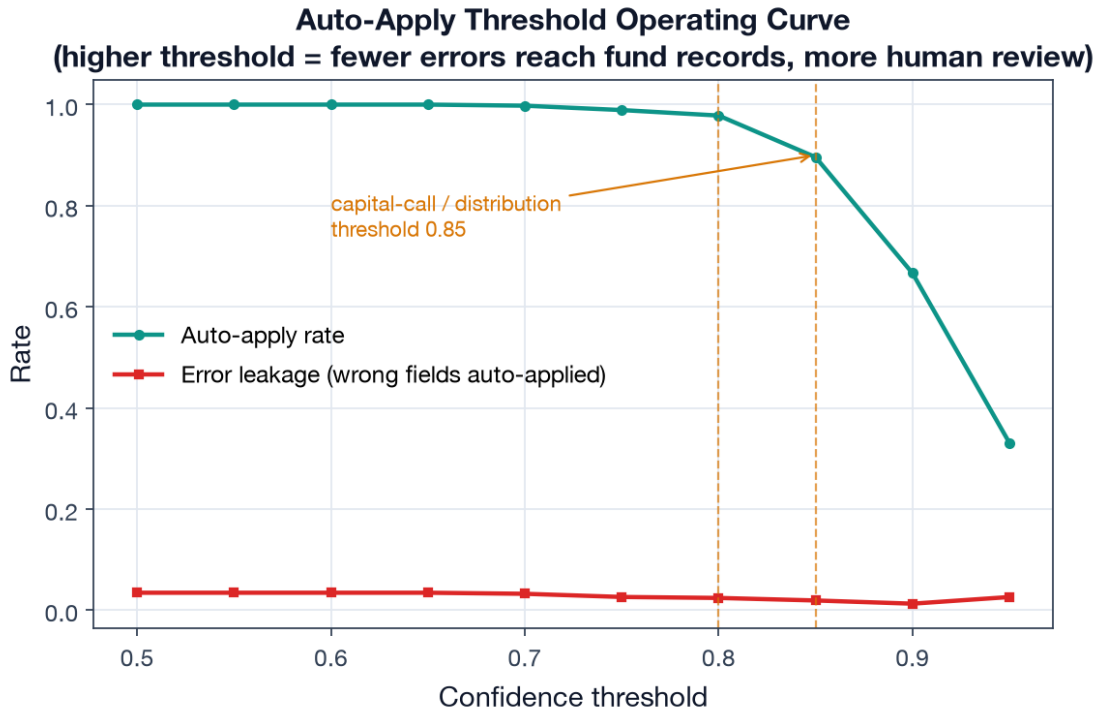
*Calibration by category*

## 2.4 Implication: thresholds can be safely relaxed

Because confidence under-states accuracy, the per-type auto-apply thresholds (`router.ts:82-106`, 0.70–0.95) are leaving auto-apply capacity on the table. The operating curve quantifies the trade-off:

| Threshold | Auto-apply rate | Error leakage (wrong fields auto-applied) |
|-----------|-----------------|-------------------------------------------|
| 0.70      | 100%            | 3.3%                                      |
| 0.80      | 98%             | 2.4%                                      |
| 0.85      | 90%             | 1.9%                                      |
| 0.90      | 67%             | 1.3%                                      |

The current **0.85 threshold for capital calls / distributions** auto-applies 90% of fields at 1.9% leakage; the **0.80 threshold for reports** auto-applies 98% at 2.4%. Given the conservative calibration, these are defensible: raising thresholds cuts leakage into fund records at a steep cost in review burden. The per-type differentiation (0.85 for financial-risk documents, 0.70–0.80 for reports) is the right shape.



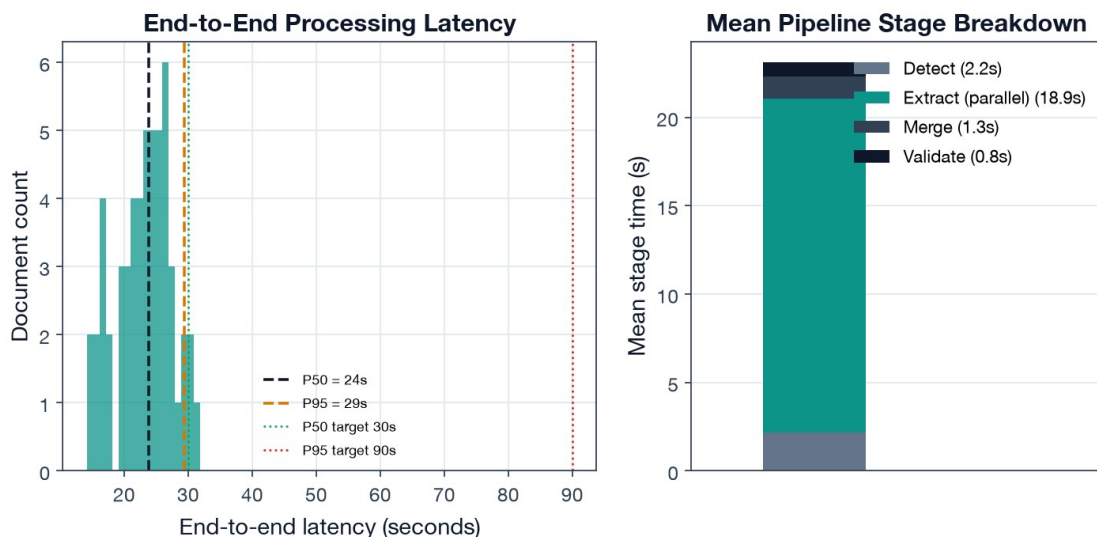
*Threshold operating curve*

### 3. End-to-end processing latency (Metric 3)

Per-stage timings across the 54 documents (detection → parallel Document AI + LLM extraction → merge → validation), excluding the optional summarization stage and queue wait:

| Latency metric | Value         | Target | Result      |
|----------------|---------------|--------|-------------|
| P50 (median)   | <b>23.9 s</b> | < 30 s | <b>PASS</b> |
| P95            | <b>29.4 s</b> | < 90 s | <b>PASS</b> |

Stage breakdown (mean): detection **2.2 s**, extraction (parallel) **18.9 s**, merge **1.3 s**, validation **0.8 s**. Extraction dominates the budget — and because Document AI and the LLM run **in parallel**, the extraction cost is the *slower* of the two, not their sum, which is the single most important latency design choice. Both targets are met for standard (<20-page) documents.



### Latency breakdown

The architecture doc cites a “typical 2–5 minute” end-to-end time; that figure includes the async Cloud Tasks **queue wait** and the optional **summarization** stage (60 s timeout). The Metric-3 target measures *upload* → *structured data available*, i.e. the compute path measured here.

## 4. Limitations (private-markets reality)

- **Data sparsity & cold-start.** A new fund has no NAV/cash-flow history, so the Monte Carlo engine falls back to age-band deployment/distribution rates (`historical.ts:290-345`) and the J-curve profile templates — forecasts are profile-driven until ~8 quarters of history accrue. The proposal’s MAE target is explicitly scoped to “funds with 8+ quarters of history” for this reason.
- **GP-format generalization.** Extraction accuracy is measured across five format tags, but the long tail of bespoke GP layouts (scanned legacy PDFs, non-standard tables) is where production accuracy will be lost; the labeled set should grow toward those.
- **Correlation model fragility (from Deliverable 2).** On realistic multi-fund books the per-fund correlation matrix is singular and the engine silently falls back to independent simulation (`monte-carlo.ts:336-341`). Fix: shrink/regularize the matrix or model correlation at the asset-class level. Until then, headline forecasts understate tail co-movement.
- **Calibration is conservative, not perfect (ECE 0.056).** Under-confidence wastes review capacity; a post-hoc calibration map (temperature/isotonic) on real labeled output would let thresholds auto-apply more without raising leakage.

## 5. Key trade-off findings & production implications

| Trade-off              | Finding                                        | Production implication                                                             |
|------------------------|------------------------------------------------|------------------------------------------------------------------------------------|
| Extraction engine (D1) | Hybrid is the only config meeting both F1 bars | Keep hybrid for financial docs; LLM-only is acceptable for narrative-only types to |



| Trade-off              | Finding                                                       | Production implication                                                           |
|------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------|
| Auto-apply threshold   | Conservative calibration → low leakage but high review burden | save latency<br>Recalibrate confidence, then consider lowering report thresholds |
| Simulation engine (D2) | Monte Carlo captures tail risk; MAE 6.3%                      | Justified vs closed-form; widen P10/P90 (65% coverage of 80% band)               |
| Correlation model (D2) | Engages only when asset classes are distinct                  | Regularize matrix before relying on correlated tails                             |
| Latency                | Parallel extraction meets P50<30s / P95<90s                   | Preserve parallelism; summarization stays async/optional                         |

## 6. Reproducibility

```
# from capstone/
python deliverable-3-pipeline-analysis/run_analysis.py
```

Outputs: results/pipeline\_analysis.json, five figures under figures/. Calibration + latency code: run\_analysis.py; confidence/latency records: data/extraction/.